D.E. SHAW INTERNSHIP INTERVIEW EXPERIENCE

About the company:

The D. E. Shaw Group (DESCO) is a global investment and technology development firm with over $60 billion in investment capital as of March 2024, and offices across North America, Europe, and Asia. Founded in 1988, the firm has earned a strong reputation for innovation, risk management, and the quality of its staff. DESCO maintains a significant presence in capital markets worldwide, investing in diverse companies and financial instruments. **D. E. Shaw India** (DESIS), established in Hyderabad in 1996, is a core member of the Global Information Technology Group, providing research, development, and support for the firm's global initiatives.

Role : Software Development Engineer in Test – 2 Month summer Internship

Stipend : 2,00,000 INR / Month

Selection Process:

1. Online Test
2. Technical Interview 1
3. HR Round
4. Technical Interview 2

Online Test:

- Coding questions: 2 (20 + 30 minutes)
- Technical MCQs: 12 (17 minutes)
- Aptitude MCQs: 14 (18 minutes)

The total test duration is 75 minutes and is conducted on the **HackerRank platform** in the campus computer center.

Coding Questions:

1. You have to select a subset of size k from an array of integers and calculate the sum of the absolute differences between all pairs of elements in that subset. Among all possible subsets of size k, you need to find the maximum possible value of this sum.

For example, consider the array [2, 3, 1, 2] and k = 3. You generate all possible subsets of size 3, namely [2,1,2], [2,3,1], [2,3,2], and [3,1,2]. For each subset, calculate the sum of absolute differences between all pairs. For [2,3,1], the sum is $|2-3|+|2-1|+|3-1|=1+1+2=4$. The maximum value across all subsets is 4.

I solved this using a backtracking approach. I generated all possible subsets to find the solution and was able to pass all three example test cases.

2. You have to construct the lexicographically smallest possible string, called post, from a given string s, using two helper strings: pre and post. You are allowed only two operations:

1. Take the first character of s and append it to the end of pre.
2. Take the last character of pre and append it to the end of post.

The goal is to carefully sequence these operations so that once all characters have been moved, the final string post is as small as possible in dictionary order. (Using a Robot to Print the Lexicographically Smallest String - LeetCode)

For example, if s = "zza", we want to form the smallest possible string. Step by step:

- Start with pre = "" and post = "".
- Move all characters from s to pre, so pre = "zza", s = "".
- Then, repeatedly move characters from the end of pre to post. First, post = "a" (moving the last character). Then post = "az". Finally, post = "azz".
- The result is "azz", which is the lexicographically smallest arrangement achievable under the given rules.

I implemented the whole logic using stack and greedy approach but could not pass the example test cases.

Technical and Aptitude Questions

The technical questions are from core subjects: DBMS, OOPS, OS, and Networks. They include sample programs for output prediction and error finding. All questions are easy to medium difficulty.

The aptitude questions are mathematical and required more effort to solve. The MCQs have negative marking (+3/-1).

Out of **511** applicants, **17** students were selected for the interview process. I was one of them.

Technical Interview 1: 1 hour

This round was primarily focused on Data Structures and Algorithms (DSA) questions.

After I introduced myself, we began the interview.

1. You are given a 2D array, distances, of size k×n, where distances[i][j] represents the distance jumped by player j in round i. In the first round, players jump in ascending order of their IDs (0, 1, 2, ..., n−1). From the second round onward, the jumping order is decided by the previous round’s results, with the player who made the longest jump going first. The winner is the player who achieves the single longest jump across all rounds. Return the ID of the winner.

Input: distances = [[5, 8, 10, 9], [6, 2, 10, 4.5], [11, 9, 5, 6]],

Output: 0

Explanation:

- Round 1 order = [0, 1, 2, 3] → jumps = [5, 8, 10, 9] → next order = [2, 3, 1, 0]
- Round 2 order = [2, 3, 1, 0] → jumps = [10, 4.5, 2, 6] → next order = [2, 0, 3, 1]
- Round 3 order = [2, 0, 3, 1] → jumps = [5, 11, 6, 9]

Maximum jumps: Player 0 = 11, Player 1 = 9, Player 2 = 10, Player 3 = 9. The winner is Player 0.

I was able to solve this problem. First, I paired the first round's distances with their respective player IDs. I tracked the maximum distance of each player in a separate array and then sorted them based on distance to determine the order for the next round. I continued this process for all subsequent rounds.

2. You are given a mountain array arr (which is strictly increasing to a peak, then strictly decreasing).

Find the peak element in the array. Your solution must run in O(logn) time complexity. (Peak index in a mountain array - Leetcode)

Input: arr = [1, 3, 5, 7, 6, 2]

Output: 7

I initially provided a linear solution, then explained the binary search approach. Interviewers then asked me to write the binary search code on paper.

3. Given a string of words and a maximum width (maxWidth), you need to write code to format the text such that each line is fully justified. (Text justification - Leetcode)

I first provided a solution, and then they continuously added more constraints and different edge cases. This went on for quite some time, with additional constraints, modifications, and logic being added to the code. In the end, I was able to satisfy them with my solution.

4. You are given an employee table with the following columns: empID (primary key), ManagerID. You need to write a query that outputs every pair of employees who have the same manager.

I wrote the query using a self-join.

They then moved on to some general computer-related questions. The first question was, "What is a CPU?" I answered this. The next question was, "Why does a computer system become slow or less efficient after long use?" I discussed both general and technical reasons with them.

Finally, they asked for a brief introduction in the last few minutes of the interview.

HR Round: 25 Minutes

This round primarily tested my interest in the job and my project management skills. The questions were related to testing and project management.

1. What is your interest in software?
2. What is testing?
3. If we offered you two opportunities, one in software development and one in testing, which would you choose?
4. What is your experience with testing?
5. Tell me about your projects.
6. What was your role in your projects?
7. How would you handle your team if you were a leader?
8. How would you handle a slow learner or a beginner on your team?

After I answered these questions, HR allowed me to ask some of my own. I asked two questions about the company and the internship.

Technical Round 2: 1 hour 10 minutes

This interview began with my introduction. The interviewers asked me some questions from my resume about programming languages.

1. What is the difference between a process and a thread?
I answered with the definitions and provided some examples.

2. Design a lift system. There are two lifts, and you have to design it in such a way that it can be handled by a single button on each floor. They wanted me to come up with a system and different test cases.

I gave a priority queue-based lift system and provided some test cases. They asked me to draw a flowchart for the system's workflow and posed a specific test case to see if my system could handle it. I confirmed that my solution would pass the test.
Finally, they asked me to write three more test cases for the same system. After a few minutes of thinking, I provided two. Then, we moved on to the next question.

3. You are given a 2D grid (array) where each cell contains an integer. Your goal is to find the maximum path sum when traveling from the first column to the last column. You can start from any cell in the first column and move to any cell in the last column. ([Maximum path sum in matrix - GeeksforGeeks])

The allowed movements are:
Move directly to the cell to the right, Move diagonally up-right and Move diagonally down-right.

Input Format: [[3, 1, 2], [6, 5, 4], [7, 8, 9]]

Output: 24 (7 -> 8 -> 9)

I gave a recursive solution and discussed its space and time complexity. I was then asked to optimize. After a tip from the interviewer, I was able to give the DP solution. I was asked to write the code for both the recursive and the dynamic programming (DP) solutions on paper.

4. You are given two arrays: a) Weights: An array where each element represents the weight of a stone. b) Directions: An array where each element represents the direction in which the stone is moving ('R' for right and 'L' for left).
When stones collide, the stone with the larger weight remains while the lighter stone is destroyed. If both stones have the same weight, both are destroyed. Stones continue to collide and be destroyed until no more collisions occur. Your task is to write a function that returns the indices of the stones that remain after all collisions have been resolved. (Asteroid Collision - LeetCode)

Input Format:

Weights: [10, 5, 8, 6, 3]

Directions: ['R', 'R', 'L', 'L', 'R']

Output: [0, 2]

I had not solved the asteroid collision problem before. I came up with an $O(n^2)$ solution that involved a loop to check all adjacent nodes for collisions, repeating this process until no more collisions occurred. I was then asked to write the code for this. When I asked for more time to optimize, they told me that this solution was sufficient.

Finally, they allowed me to ask some questions. I asked if my explanation was clear or if I needed to improve, and I also asked for their advice on a future career in the software industry.

Then the results came out. Out of the **17** students, **3** students were selected for a summer internship, including me. I was from MIT, and the other two students were from CEG.

After completing our two-month internship, we all received **Pre-Placement Offer** (PPO) from the company. (I'm writing this after the internship has ended.)

General Tips:

- Be consistent and stay positive.
- You don't have to be perfect - just keep trying.
- Struggles and self-doubt are completely normal parts of the learning process.
- Learn to communicate clearly.
- LeetCode is great for practicing competitive coding, and HackerRank is, too.
- NeetCode.io and Striver are free gems for preparation. (Many other resources are available; choose the one that best suits you.)
- Revise your core concepts regularly, and practice aptitude questions.

“You don’t have to be selected to prove you’re the best. Just give your best, be proud of your effort, and trust that hard work always pays off. All the best ✨”

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